

Bank Marketing Using Data Science and Machine Learning

Project Title: Bank Marketing Prediction Using Data Science and Machine Learning

Project Overview

Predict whether a customer subscribes to a bank term deposit using data science and machine learning.

Problem Statement

Build a binary classification model to predict term deposit subscription (yes/no).

Dataset

Bank Marketing dataset (45,211 rows, 17 columns). Target: y.

Workflow

Data Collection, Data Understanding, Data Cleaning, EDA, Feature Engineering, Train-Test Split, Model Training, Evaluation, Model Saving, Deployment.

Algorithms

Logistic Regression, Decision Tree, Random Forest, SVM, KNN, Naive Bayes, XGBoost (optional).

Evaluation Metrics

Accuracy, Precision, Recall, F1-score, ROC-AUC Score, Confusion Matrix.

Tools

Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, Git, GitHub, Docker.

Outcome

Predicts likely subscribers and demonstrates an end-to-end ML workflow.